

Hao Guo

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📍 Remote Sensing Building, Peking University, Beijing, China
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EDUCATION

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| Peking University <ul style="list-style-type: none">• Ph.D. in Cartography and Geographic Information Systems• Advisors: Prof. Lun Wu and Prof. Yu Liu• Dissertation: Automatic Zoning Methods for Modeling Spatial Process Heterogeneity | 2020.09–2025.06 |
| Peking University <ul style="list-style-type: none">• B.S. in Mathematics and Applied Mathematics (dual degree)• GPA: 3.75/4.0 | 2017.09–2020.06 |
| Peking University <ul style="list-style-type: none">• B.S. in Geographic Information Science• GPA: 3.85/4.0, ranked 1/22• Thesis: Estimating Gravitational Attractions of Chinese Cities Using Web Search Index Data | 2016.09–2020.06 |

EXPERIENCE

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| Postdoctoral Fellow
Institute of Remote Sensing and Geographic Information Systems, Peking University <ul style="list-style-type: none">• Statistical inference for spatial networks• Symbolic regression and geographic knowledge discovery | 2025.09–Present |
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TEACHING

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| Teaching Assistant
Introduction to Geographic Information Systems, Peking University <ul style="list-style-type: none">• Taught basic QGIS usage. | Spring 2021 |
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RESEARCH INTERESTS

- Spatial data analytics
- Spatial statistics
- Spatial optimization
- Geospatial artificial intelligence

PUBLICATIONS

Equal contribution

- [1] **Hao Guo**, Andre Python, Yu Liu. Extending regionalization algorithms to explore spatial process heterogeneity. *International Journal of Geographical Information Science*, 2023. [doi:10.1080/13658816.2023.2266493](https://doi.org/10.1080/13658816.2023.2266493)
- [2] **Hao Guo**, Han Wang, Di Zhu, Lun Wu, A. Stewart Fotheringham, Yu Liu. RegionGCN: Spatial-Heterogeneity-Aware Graph Convolutional Networks. *Annals of the American Association of Geographers*, 2026. [doi:10.1080/24694452.2025.2558661](https://doi.org/10.1080/24694452.2025.2558661)
- [3] **Hao Guo**, Lei Dong, Lun Wu, Yu Liu. Tackling spatial heterogeneity in geographical analysis: An overview. *Acta Geographica Sinica*, 2025. [doi:10.11821/dlxb202503001](https://doi.org/10.11821/dlxb202503001)
- [4] **Hao Guo**#, Weiyu Zhang#, Haode Du#, Chaogui Kang, Yu Liu. Understanding China's urban system evolution from web search index data. *EPJ Data Science*, 2022. [doi:10.1140/epjds/s13688-022-00332-y](https://doi.org/10.1140/epjds/s13688-022-00332-y)
- [5] **Hao Guo**#, Weiyu Zhang#, Junjie Yang, Yuanqiao Hou, Lei Dong, Yu Liu. Distilling human mobility models with symbolic regression. *Geographical Analysis*, 2026. [doi:10.1111/gean.70043](https://doi.org/10.1111/gean.70043)

- [6] **Hao Guo**, Lei Dong, Xiaodong Zhang, Yu Liu. Regional gravity: a zoning heuristic for detecting heterogeneous distance decay effect in spatial interaction. *International Journal of Geographical Information Science*, accepted.
- [7] Yu Liu, **Hao Guo**, Haifeng Li, Weihua Dong, Tao Pei. A note on GeoAI from the perspective of geographical laws. *Acta Geodaetica et Cartographica Sinica*, 2022. doi:10.11947/j.AGCS.2022.20220125
- [8] Tianyou Cheng, **Hao Guo**, Xiao-Jian Chen, Quanhua Dong, Chaogui Kang, Yu Liu. Constructing topology-constrained distance cartograms with application on spatial interaction data. *Transactions in GIS*, 2024. doi:10.1111/tgis.13168
- [9] Chengling Tang, Lei Dong, **Hao Guo**, Xuechen Wang, Xiao-Jian Chen, Quanhua Dong, Yu Liu. Downscaling spatial interaction with socioeconomic attributes. *EPJ Data Science*, 2024. doi:10.1140/epjds/s13688-024-00487-w
- [10] Han Wang, Zhou Huang, **Hao Guo**, Ganmin Yin, Yi Bao, Xiao Zhou, Yong Gao, Linna Li. GWRBoost: A geographically weighted gradient boosting framework for enhanced explainable quantification of spatially varying relationships. *Annals of the American Association of Geographers*, 2026. doi:10.1080/24694452.2026.2648327
- [11] Han Wang, Xiao Zhou, **Hao Guo**, Quanhua Dong, Zhou Huang. Quantifying spatially-varying impacts of public transport on NO2 concentrations with big geo-data. *Environmental Monitoring and Assessment*, 2023. doi:10.1007/s10661-023-11289-4
- [12] Yi Wang#, Yizhi Zhang#, Quanhua Dong, **Hao Guo**, Yingchun Tao, Fan Zhang. A multi-view graph neural network for building age prediction. *ISPRS Journal of Photogrammetry and Remote Sensing*, 2024. doi:10.1016/j.isprsjprs.2024.10.011
- [13] Yu Liu, Keli Wang, Xiaoyue Xing, **Hao Guo**, Weiyu Zhang, Qinyao Luo, Song Gao, Zhou Huang, Haifeng Li, Xin Li, Jiaoe Wang, Jinfeng Wang, Di Zhu. Spatial effects in geographic analysis. *Acta Geographica Sinica*, 2023. doi:10.11821/dlxb202303001

TALKS

- **Hao Guo**. Regional gravity models of spatial interaction. 20th Chinese Conference on Geographic Information Science Theory and Methods (invited talk), Xiamen, Nov 2025.
- **Hao Guo**, Junlong Zhang, Junjie Yang, Lei Dong. On commuting duration and the daily travel time budget. NetMob 2025 Data challenge track (oral), Conservatoire National des Arts et Métiers, Paris, Oct 2025.
- Yu Liu, **Hao Guo**. Spatial-heterogeneity-aware geographic regression. Annual Conference of Geographic Models and Analytics Speciality Group, Chinese Society of Geographic Information Science (invited talk), Beijing, Aug 2024 .
- **Hao Guo**. Distilling geographic laws from human mobility data. Urban Data, Science and Technology Workshop, Tsinghua University, Beijing, Jun 2024.
- Yu Liu, **Hao Guo**. A spatial clustering and regionalized regression framework for spatial knowledge discovery. 17th Chinese Conference on Geographic Information Science Theory and Methods, Hangzhou, Oct 2021 (invited talk).

GRANTS

- National Natural Science Foundation of China, General Fund (01/2027-12/2030): Generalized Generation of City-wide Individual Trajectories Based on Mobility Data of Local Area Residents (Collaborator) | RMB 430,000

SERVICE

- Journal reviewer: *Annals of the American Association of Geographers*, *Geographical Analysis*, *IEEE TKDE*, *IJGIS*, *Transactions in GIS*, *Urban Climate*, *Urban Informatics*, *Acta Geodaetica et Cartographica Sinica*.
- Conference reviewer: *ACM SIGSPATIAL GeoAI*.

AWARDS AND HONORS

Academic awards

- Merit Student of Peking University, 2017, 2018, 2019, 2024
- Luo Yuehua Scholarship, 2024
- Study Excellence Award of Peking University, 2021
- Special Academic Scholarship of Peking University, 2021
- Excellent Graduate of Peking University, 2020
- May Fourth Scholarship (Highest Honor Scholarship of Peking University), 2019

- National Scholarship, 2018
- Wei-Lin Scholarship, 2017

Competition awards

- Finalist, Alibaba Global Mathematics Competition (approximately top 1.5%), 2022, 2023, 2024
- Provincial First prize, National College Mathematics Competition (non-mathematics major), 2017, 2018
- Finalist, Mathematical Contest in Modeling (MCM), 2018
- Excellence award, Peking University Programming Competition, 2018
- First prize, National High School Mathematics Competition, 2015
- Silver medal, Southeastern Mathematical Olympiad of China (11th Grade Group), 2014

SKILLS

- Computer: CCF CSP score 400 (top 1%); proficient in Python, C++, and R; experienced with $\text{\LaTeX}$.
- English: CET-6 score 650; good spoken English.
- Interests: Orienteering, logical puzzles, fiction writing.